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Statistical Models

Week 7: Models for over- and underdispersed counts, and model validation by AIC

comments

The main topic of this week are slightly advanced models for count data, and model assessment. As time allows, we will also treat some "loose ends", such as the Fisher scoring method behind the `glm` function, and more background on contingency tables and the Poisson trick.

The Poisson model is a good start for count data. Sometimes, however, there is more variation in the data than we would expect under a Poisson model. This is called *overdispersion*. One reason for overdispersion is that the population is composed of subpopulations with different mean. We will learn about the negative binomial distribution as a mixture of Poisson distributions. Another reason for overdispersion is *zero-inflation*: there are more "0"s in the data than we would expect under the models in question.

A very popular general tool for assessment of model fit is *Akaike's information criterion (AIC)*. You will apply this to glms, and learn about its theoretical background: it is related to the mean Kullback-Leibler divergence between the "true" distribution and all fitted distributions from a family.

Relevant Readings

- Models for overdispersed count data, zero-inflation: Agresti, 7.3 and 7.4.1, also start of Section 10.6 in Davison (without the paragraph on quasi-likelihood).
- Akaikes information criterion (AIC): Davison, 4.7

Lecture Material

- Link to onenotes on Google drive
- Simulation of a mixed Poisson distribution

Exercises 7

The first two problems are data analysis problems. If you have only little time, attempt at least one of them. The third and fourth problem add to your statistician's toolbox.

Problem 7.1: Lizards: compare modelling approaches

Davison Practical 10.5:
Data set `lizards` contains the numbers of two species of lizard, *grahami* and *opalinus*, observed in Whitehouse, Jamaica, at two different heights, two perch diameters, in sun or shade, at three times of the day. The data were collected by observing occupied perches and recording the appropriate height, perch diameter, and so forth.
Interest lies in the difference in habitat choice between the species. Analyze the data by approaches (i)-(iii) below, comparing their relative advantages and disadvantages, and paying particular attention to the interpretation of the models you choose.

i. A conventional analysis of variance.

ii. A generalized linear model with binomial error, based on the probability that a perch is occupied by one species, given that it is occupied. Discuss the similarities and dissimilarties with (i).
iii. A log-linear model.

(Sections 10.4–10.5; Bishop et al., 1975, p. 164)

Problem 7.2: Homicides

(based on Agresti, problem 7.31)
Table 7.5. in Agresti(2015), shown below, summarizes responses of 1308 subjects to the question: within the past 12 months, how many people have you known personally that were victims of homicide?
The table shows responses by race, for those who identified their race as white or as black. You can download the data `homicides.csv` here or find it in the`data` folder on Google Drive.

Table 7.5 Number of Victims of Murder Known in the Past Year, by Race, with Fit of Poisson and Negative Binomial Models

	Data		Poisson GLM		Negative Binomial GLM	
Response	Black	White	Black	White	Black	White
0	119	1070	94.3	1047.7	122.8	1064.9
1	16	60	49.2	96.7	17.9	67.5
2	12	14	12.9	4.5	7.8	12.7
3	7	4	2.2	0.1	4.1	2.9
4	3	0	0.3	0.0	2.4	0.7
5	2	0	0.0	0.0	1.4	0.2
6	0	1	0.0	0.0	0.9	0.1

Source: 1990 General Social Survey, file `Homicide.dat` at `www.stat.ufl.edu/~aa/glm/data`.

a. Let y_i denote the response for subject i and let $x_i = 1$ for blacks and $x_i = 0$ for whites. Fit the Poisson GLM $\log \mu_i = \beta_0 + \beta_1 x_i$ and interpret $\hat{\beta}$.

b. Describe factors of heterogeneity such that a Poisson GLM may be inadequate. Fit the corresponding negative binomial GLM, and estimate how the variance depends on the mean. What evidence does this model fit provide that the Poisson GLM had overdispersion? (Table 7.5 also shows the fits for these two models.)

   R hint: use `glm.nb` from package `MASS`

c. Verify Table 7.5 (calculate the numbers in the table yourself). Hints: get the predicted numbers from the probabilities under the fitted distributions (there were 159 black and 1149 white persons in the study).

   R hints: obtain negative binomial probabilities with `dnbinom`. For frequencies of the actual data, use the R function `table`.

d. Show that the Wald 95% confidence interval for the ratio of means for blacks and whites is (4.2, 7.5) for the Poisson GLM but (3.5, 9.0) for the negative binomial GLM. Which do you think is more reliable? Why?

Problem 7.3: The index of dispersion test

For i.i.d. samples $Y = (Y_1, \ldots, Y_n)$, the *index of dispersion* is defined as

$$D(Y) = \frac{S^2}{\overline{Y}}, \text{ where } \overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i, \quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2.$$

If the sample comes from a Poisson distribution, the expectation of the numerator is the same as the expectation of the denominator. Therefore, values close to 1 suggest Poisson distribution, while values larger than 1 indicate overdispersion and values less than 1 indicate underdispersion.
Under the Poisson assumption, $(n-1)D(Y)$ is approximately χ^2 distributed with $n-1$ degrees of freedom. This can be used to construct an approximate test for the hypothesis that the data stem from a Poisson distribution. A popular rule of thumb tells that the approximation is good, when the mean of Y is larger than 5 (in the literature, this is commonly called $m > 5$), even when the sample size n is small.

a. The rule of thumb is apparently due to Hoel (1943)[a], who studied the quality of the χ^2 approximation by numerical calculations. He looked at various combination of theoretical mean m and sample size n, and concluded that the test becomes slightly conservative if $m \leq 2$.

   Conduct a simulation experiment to check Hoel's findings: estimate the true rejection rate of a χ^2 based index of dispersion test, for $m \in \{1, 2, 5, 25\}$ and $n \in \{3, 5, 10, 25\}$.

   Hints

   - to estimate the rejection rate, find the critical values of the χ^2 distribution. For a two-sided test that is sensitive to both over- and underdispersion, reject the null hypothesis if D is smaller than the $\alpha/2$- quantile or larger than the $(1-\alpha/2)$-quantile of the χ^2 distribution.
   - An R hint for fast generation of (`nrep`) realizations of D:

```
Y <- matrix(rpois(n * nrep, m), nrow = n)
D <- apply(Y, 2, var) / apply(Y, 2, mean)
```

b. Use the index of dispersion test to check if the Poisson model is appropriate for the data from the previous exercise about homicides (Agresti, Table 7.5). Treat black and white persons as separate samples from possibly different Poisson distributions.

[a] Paul G. Hoel (1943): On Indices of Dispersion. *The Annals of Mathematical Statistics*, **14**(2): 155–162.

Problem 7.4: Simulation based model tests

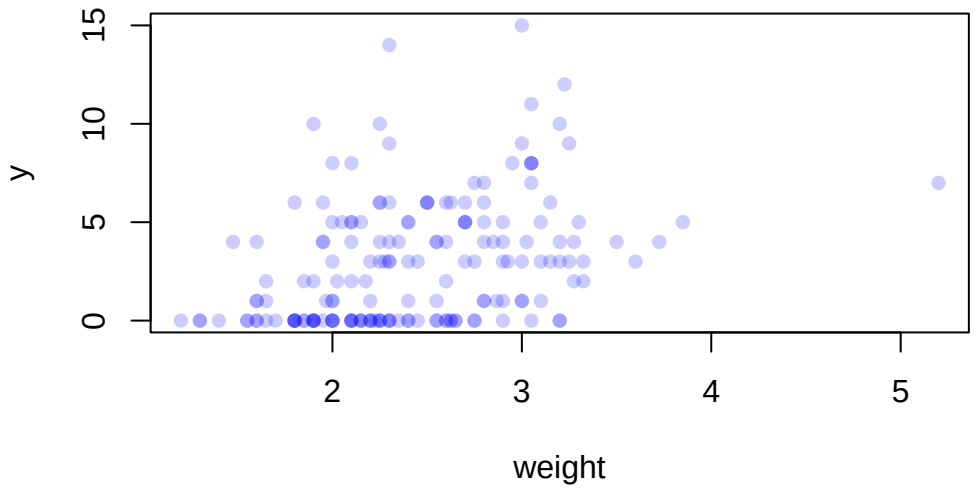
a. Consider Exercise 6.3, "A graphical model check" from last week. You can turn this "gut feeling" into a test with well defined rejection rate $\alpha = 0.05$ by simulating 19 realizations, and arranging them on a grid together with the one observed plot at a random position within the 20 plots. Then ask an impartial judge (e.g. your grandmother) to point out the most unusual plot – this test rejects the null hypothesis if the judge pointed at the plot made from the actual data. One problem here is that the test may be conservative because you test the fitted model. But for a given model, the size of this test[a] is equal to α.

   Discuss why this is a test of size α for a given model. What is α if you only simulate 9 plots?

R hint

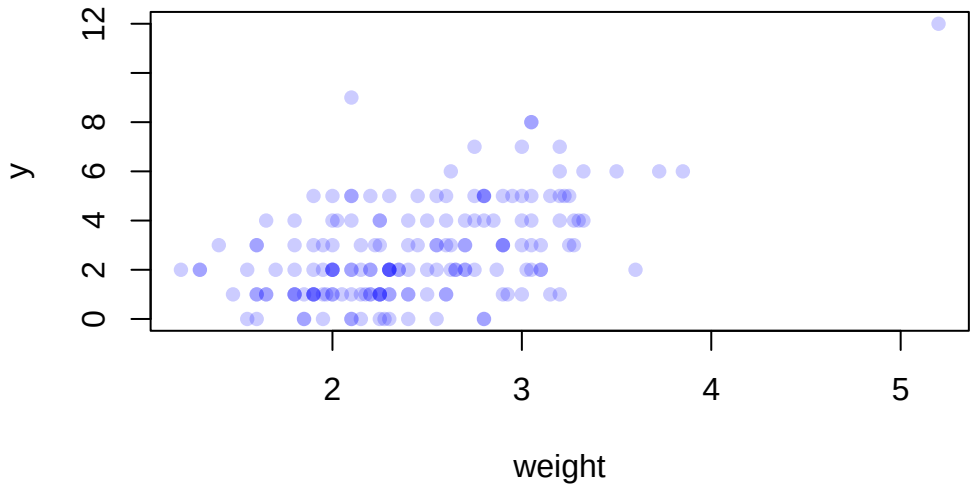
In Problem 6.3 you were asked to compare scatterplots of the crabs data with scatterplots of simulated data from the fitted model. Here is one way to solve this.
We make the scatterplot with transparent dots to make multiplicity of observations visible. `adjustcolor` is an easy way to define transparent colors

```
crabs <- read.csv("./data/crabs.csv")
transparentblue <- adjustcolor("blue",  alpha = 0.2)
plot(y ~ weight, data = crabs, pch = 16, col = transparentblue)
```



The R function `simulate` simulates response values from a fitted model. It returns a `data.frame` where each column corresponds to one simulation. If you do not specify the argument `nsim`, it will just simulate one realization.

```
# fit model
fitted <- glm(y ~ weight, data= crabs, family = poisson)
# make 4 simulations from fitted model and plot
new_y <- simulate(fitted)
plot(new_y[,1] ~ weight, data = crabs, pch = 16, col = transparentblue, ylab = "y")
```



b. Check out the R package `DHARMa`. Explain what it does, discuss if you find it useful :-)

[a] size of a test = true rejection rate

I will probably use the last problem as an example in one of the lectures this week. Watch out for some R goodies :-)

Problem 7.5: Shark attacks

(modified from Agresti, problem 7.30)
A headline in *The Gainesville Sun* (February 17, 2014) proclaimed a worrisome spike in shark attacks in the previous 2 years. The reported total number of shark attacks in Florida per year from 2001 to 2013 were 33, 29, 29, 12, 17, 21, 31, 28, 19, 14, 11, 26, 23.

a. Are these counts consistent with a null Poisson model or a null negative binomial model [a]?
   Test the Poisson model against the negative binomial alternative.

b. Analyze the evidence of a positive linear trend over time, both for a Poisson and a negative binomial model. Comment on the results.

Hints

- Here are the data for copy-paste:

```
attacks <- c(33, 29, 29, 12, 17, 21, 31, 28, 19, 14, 11, 26, 23)
year <- 2001 : 2013
```

- You can use `glm.nb` from package `MASS` to fit negative binomial glms. You can call this function as `MASS::glm.nb`, then you don't need `library("MASS")`.
- The Poisson model can be seen as a submodel of the negative binomial model. Therefore, you can use a likelihood-ratio test. To this end, R-function `logLik` may be handy. Check out how to use it.
- Function `summary` for fitted linear models shows p-values for the coefficients. This can be used in part b.

[a] "null model": mean (and variance) do not depend on any explanatory variables